

Linear Algebra for Data Science

A branch of mathematics essential for understanding data science, machine learning, and AI. This course covers the core concepts every data science practitioner needs.

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Scalars & Vectors

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Matrix Operations

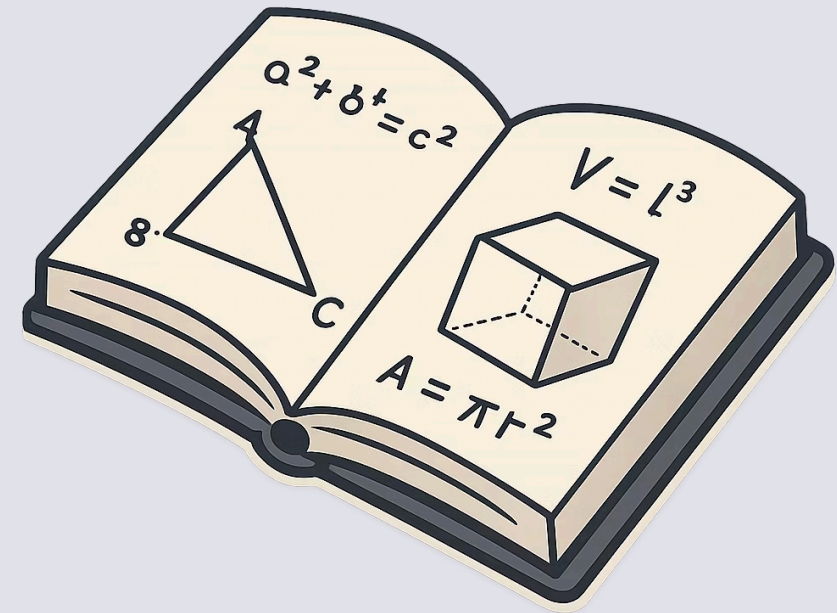
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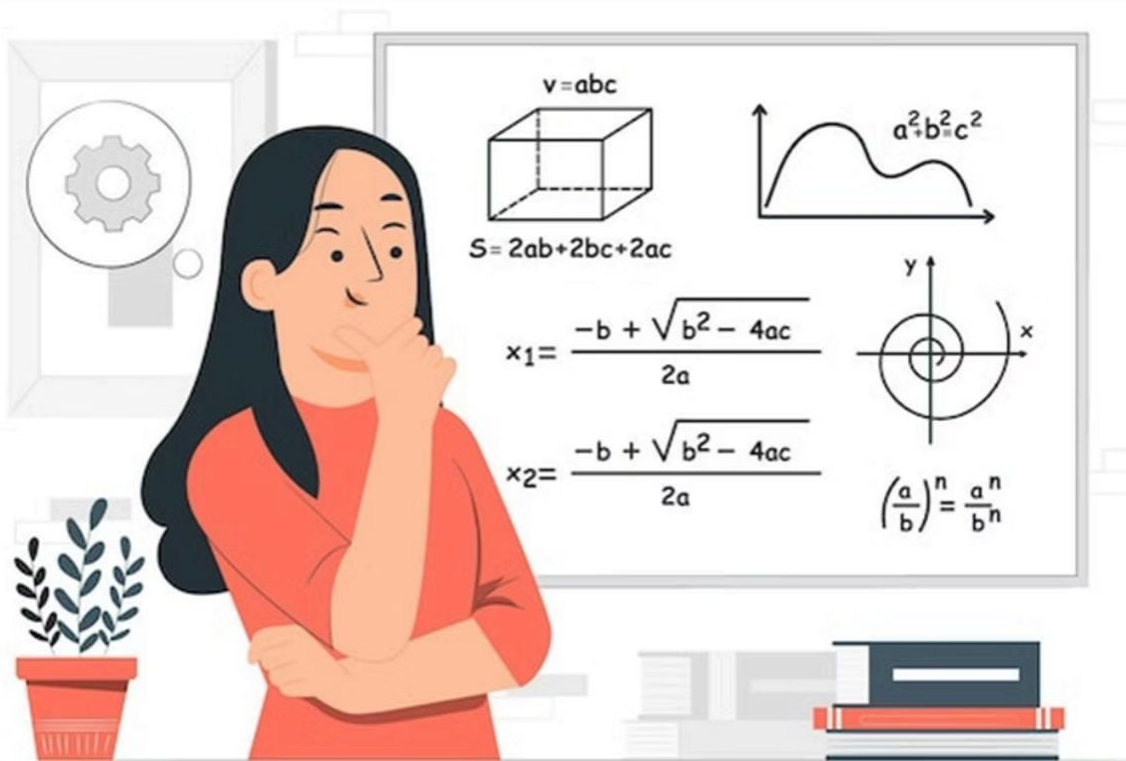
Determinants & Inverses

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Eigenvalues & Eigenvectors

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Introduction: What Is Linear Algebra?

Linear algebra uses **matrices and vectors** to represent and simplify large amounts of data. It is the foundation for machine learning tasks like reducing data dimensions and selecting optimal hyperparameters.

Notation

Clear notation simplifies reading algorithms in papers, books, and Python code.

Operations

Key operations: **Addition, Multiplication, Inversion, Transposition**

Representation

Data is represented using linear equations organized into matrices and vectors.

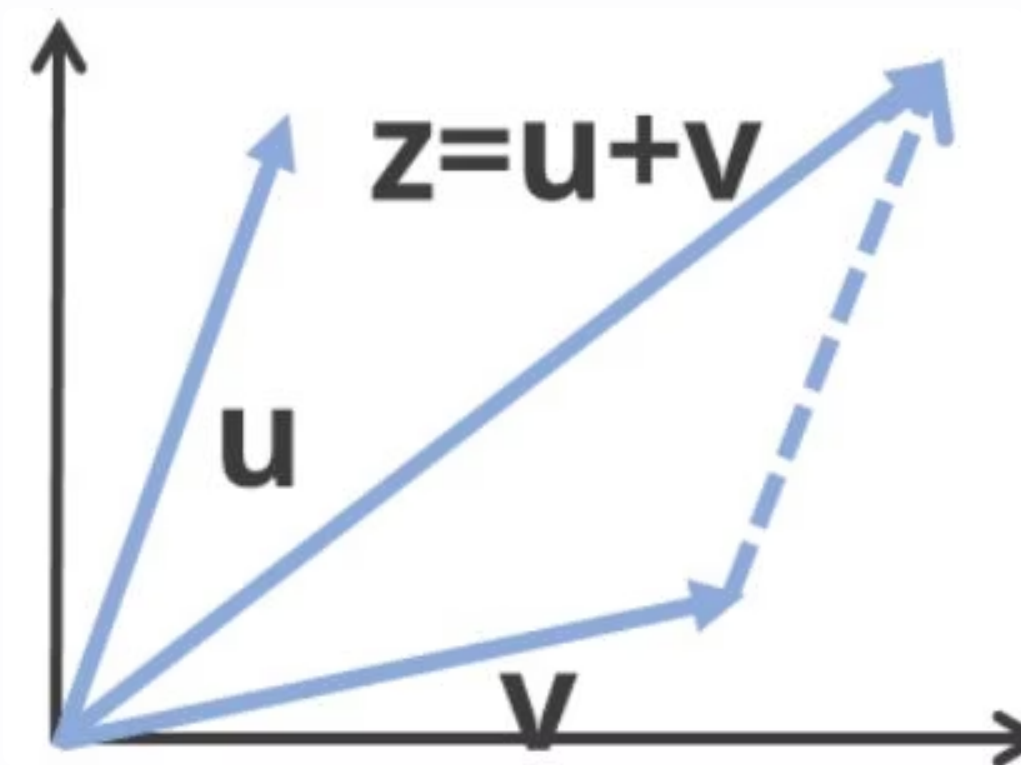
Scalars, Vectors & Vector Operations

Scalars vs. Vectors

A **scalar** has only magnitude (e.g., length, area). A **vector** has both magnitude and direction (e.g., velocity, force).

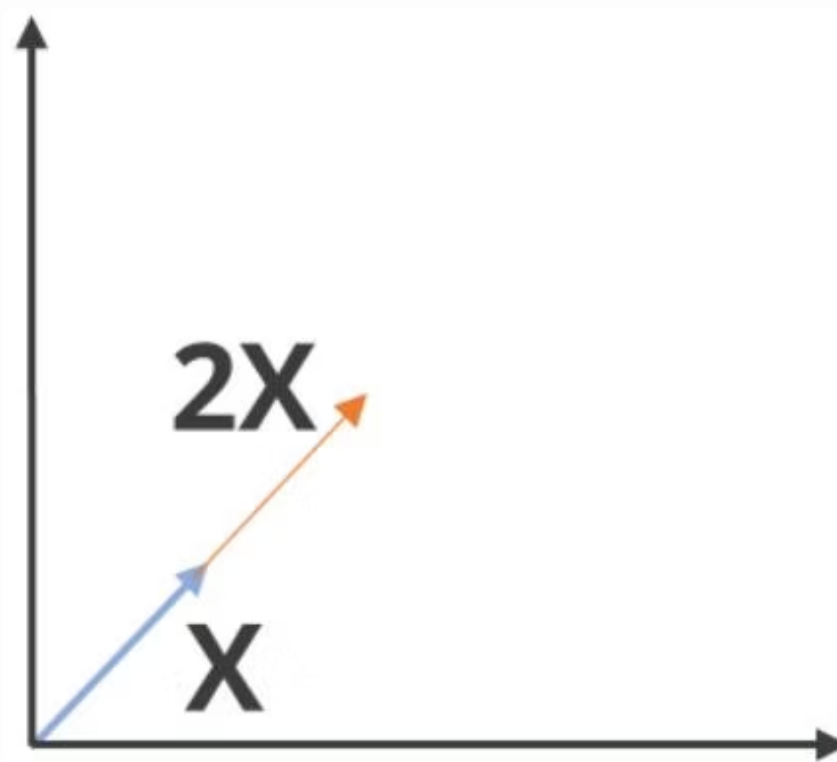
Vectors are the most fundamental mathematical objects in data science – ordered lists of finite numbers, represented as bold $\mathbf{x}$ or $\vec{x}$.

Vector Addition



$\mathbf{z} = \mathbf{u} + \mathbf{v}$ – vectors add tip-to-tail.

Scalar Multiplication



Multiplying vector $\mathbf{X}$ by 2 gives $2\mathbf{X}$ – same direction, double length.

□ Dot Product: $\mathbf{u} \cdot \mathbf{v} = x_1y_1 + x_2y_2 + x_3y_3$ | Cross Product: $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta$

Matrices: Operations & Key Concepts

A **matrix** is a rectangular array of numbers arranged in rows and columns. Matrices must have matching dimensions for addition/subtraction; for multiplication, columns of the first must equal rows of the second.

1

Addition / Subtraction

Element-wise: $X + Y = [a_{ij} + b_{ij}]$. Matrices must have the same order.

2

Scalar Multiplication

Every element is multiplied by scalar C : $CX = [ca_{ij}]$

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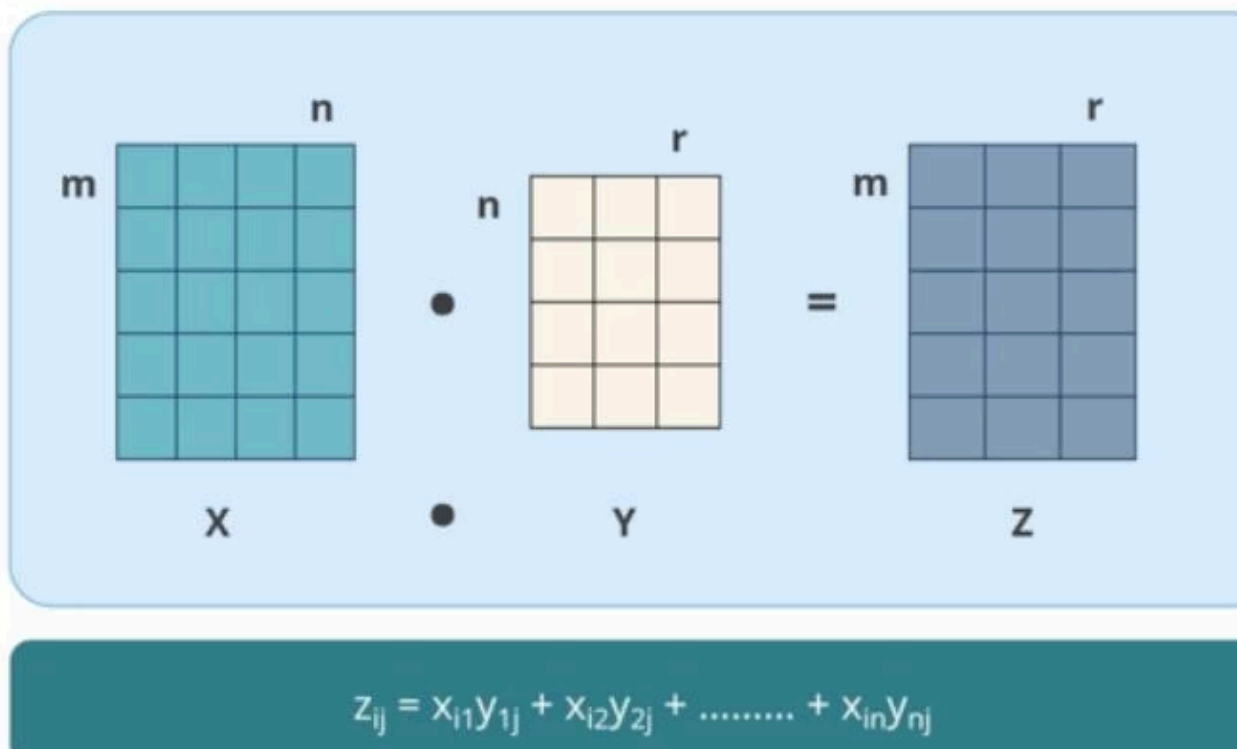
Matrix Multiplication

$Z_{ij} = X_{i1}Y_{1j} + \dots + X_{in}Y_{nj}$. Result of $m \times n$ and $n \times r$ is $m \times r$.

4

Transpose

Swap rows and columns: a 2×3 matrix becomes 3×2 , denoted X^T .



Rank, Determinant, Inverse & Eigenvalues

Rank of a Matrix

The maximum number of linearly independent rows or columns. Convert to **row echelon form** and count non-zero rows. Example: a matrix with 2 non-zero rows has rank 2.

Determinant & Identity

The determinant is a scalar value defined only for square matrices:
 $|X| = ad - bc$ (2×2). The **identity matrix** I has 1s on the diagonal and 0s elsewhere – equivalent to the number 1: $XI = IX = X$.

Inverse of a Matrix

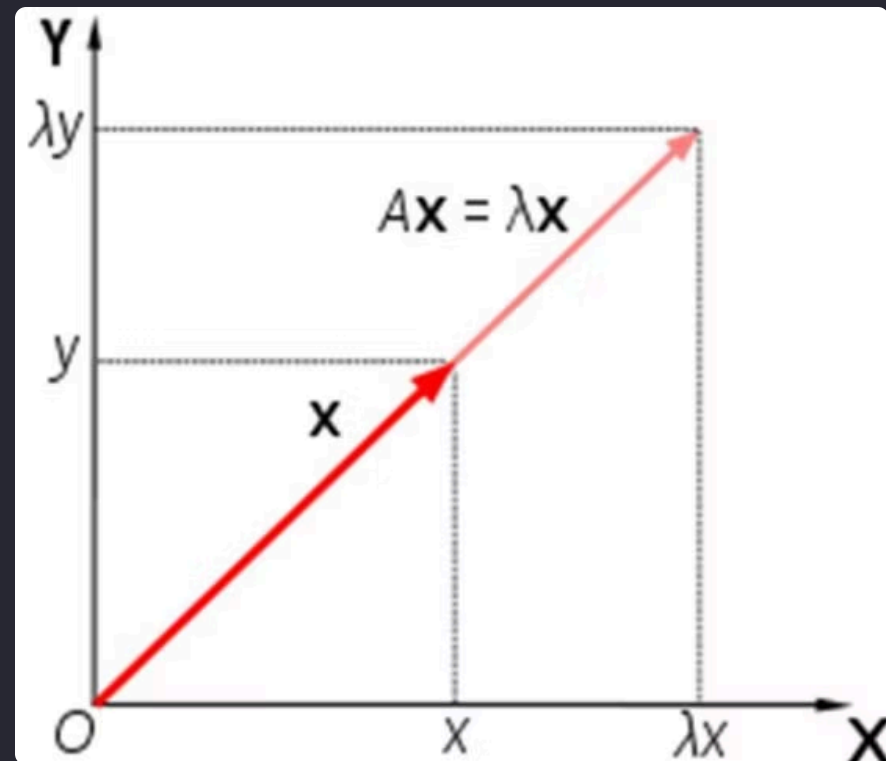
$X \cdot X^{-1} = I$. The inverse exists only if the determinant is non-zero.
Computed using cofactors and the adjoint matrix.

Eigenvalues & Eigenvectors

For matrix A , scalar λ is an **eigenvalue** and vector X is an **eigenvector** if:

$$AX = \lambda X$$

Eigenvectors do **not change direction** under transformation – they only stretch or compress. Eigenvalues measure that scaling.



Used in: linear transformations, differential equations, and structural analysis.